

STUDENT ACTIVITY

Scenario-Based Use Case Design — Activity

Activity: Data Preprocessing and Feature Strategy Design

Q: Read the business narrative below and design three distinct Data Preprocessing frameworks (each explicitly defining the Technique, Purpose, and Best Use Case) based on the provided text:

- One framework optimized for the computer vision team.
- One framework optimized for the predictive analytics team.
- One framework optimized for the data science team.

Next, based on the provided text, describe why applying a Train-Test Split is a critical final step before training these models in three sentences. Ensure you explicitly mention the common data split ratio and how it impacts overfitting and model evaluation.

The Narrative:

A healthcare analytics startup is standardizing its data pipeline for three new AI initiatives. The computer vision team is building a Deep Learning model that processes X-ray images, requiring the numerical data to be scaled into a fixed range. The predictive analytics team is training a Logistic Regression model using patient vitals that follow a strictly normal distribution. Finally, the data science team is struggling with a massive dataset containing hundreds of irrelevant columns, which is negatively impacting model simplicity and training speed. The lead architect requires a specific preprocessing technique for each team to ensure algorithm efficiency before they divide their final datasets for model evaluation.

Activity: Data Preprocessing and Feature Strategy Design

Data Preprocessing Framework for the Computer Vision Team

Technique	Purpose	Best Use Case
Normalization (Min-Max Scaling)	Scales numerical values into a fixed range (usually 0 to 1) to improve model performance and convergence.	Best for Computer Vision and Deep Learning models, such as processing X-ray images , where pixel values need to be normalized.

Data Preprocessing Framework for the Predictive Analytics Team

Technique	Purpose	Best Use Case
Standardization (Z-Score Scaling)	Transforms numerical data to have a mean of 0 and a standard deviation of 1 , ensuring consistent feature scales.	Best for Predictive Analytics models such as Logistic Regression , especially when the data follows a normal distribution .

Data Preprocessing Framework for the Data Science Team

Technique	Purpose	Best Use Case
Feature Selection	Removes irrelevant or unnecessary features to reduce dataset complexity, improve training speed, and enhance model performance.	Best for large datasets with many irrelevant columns, improving efficiency, simplicity, and reducing overfitting.

Importance of Train-Test Split

A **Train-Test Split** is the final preprocessing step that divides the dataset into separate **training** and **testing** sets before model training. A common split ratio is **80% for training and 20% for testing**, allowing the model to learn from one portion of the data and be evaluated on unseen data. This process helps **reduce overfitting**, provides a fair evaluation of model performance, and ensures that the model generalizes well to new, real-world data.

Data Preprocessing Frameworks

Team	Technique	Purpose	Best Use Case
Computer Vision Team	Normalization (Min-Max Scaling)	Scales data into a fixed range (0–1) for efficient learning.	Deep Learning models processing X-ray images .
Predictive Analytics Team	Standardization (Z-Score Scaling)	Converts data to a mean of 0 and standard deviation of 1.	Logistic Regression with normally distributed patient vitals.
Data Science Team	Feature Selection	Removes irrelevant columns to reduce complexity and improve training speed.	Large datasets with hundreds of unnecessary features.

Why is Train-Test Split Important?

The **Train-Test Split** divides the dataset into **training** and **testing** sets before model training. The most common split ratio is **80% training and 20% testing**. It helps **prevent overfitting**, enables reliable **model evaluation**, and ensures the model performs well on unseen data.